

23. (2.25 points)

a. (1 point)

Describe two drawbacks of using default avoidance as the reference point for setting capital requirements in ERM.

b. (0.5 point)

Briefly describe two other meaningful reference points for setting capital requirements.

c. (0.75 point)

Using one of the reference points identified in part b. above, develop a minimum capital requirement that relates a maximum capital loss tolerance to a TV@R measurement.

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strategic risk present in business decision making.

QUESTION 23	
TOTAL POINT VALUE: 2.25	LEARNING OBJECTIVE: C2: Describe the rationale for, methods for, and effect of managing insurance and financial risks.
SAMPLE ANSWERS	
Part a: 1 point	
<i>Sample Answer 1</i> Default is a very unlikely outcome in the far tail of the distribution of outcomes. The ERM model is probably not very accurate at this point in the distribution so using default avoidance as a reference point may not yield accurate results from the ERM model. Default avoidance mainly protects policyholder. However, other stakeholders (e.g., Shareholders) may care about large partial decreases in capital. To protect all stakeholders, need to choose more likely reference point than default.	
<i>Sample Answer 2</i> This requires selecting a capital level deep in the tail of the loss distribution, which is exactly where the loss distribution is least reliable. Default avoidance mainly protects policyholders. Shareholders can be hurt at losses lower than default level. Thus a lower level than default level may be more meaningful for the firm.	
<i>Sample Answer 3</i> Shareholders are impacted by a loss in value before the company is close to default. Capital requirements should consider protecting shareholders. Default usually happens far out into the tail of a loss distribution where the results may not be as credible. Capital requirements should be based on a credible estimate.	
Part b: 0.5 point	
<i>Sample Answer 1</i> Sufficient capital to continue servicing renewals. Sufficient capital to withstand and thrive after a catastrophe.	
<i>Sample Answer 2</i> Rating agency requirement- what level of capital is required to maintain rating Point at which capital could support renewal book	
<i>Sample Answer 3</i> Setting capital at a level that maximizes franchise value Setting capital at a level to service renewal book	

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Part c: 0.75 point <i>Sample Answer 1</i> Suppose renewals are 80% of the book, so we want to minimize the chance that we will lose more than 02% of our capital in a given year. We want to set our capital = $5 \times TV@R90\%$ this means one out of 10 years were are expected to lose an amount of capital equal to $TV@R90\%$ which is $TV@R / (5 \times TV@R90\%) = 20\%$, so we can still service renewals. <i>Sample Answer 2</i> No more than 20% of capital to 1 an 100 event (needed to maintain capital to service ongoing business). Minimum Capital Requirement (MCR) = $TV@R90\% \times 5$. $TV@R90\%$ is the expected value of a 1 in 100 event. If that occurred, we would lose $TV@R90\% / (5 \times TV@R90\%) = 20\%$ of capital. <i>Sample Answer 3</i> To hold enough capital to not only survive a major CAT but thrive in its aftermath; Set minimum capital equal to 6 time 95th percentile $TV@R$. This ensures that an average 1 in 20 year event will deplete only 1/6th of the company's capital. So, even after this event the company will not just survive, but should have enough remaining capital to thrive.
EXAMINER'S REPORT This was a challenging question; however, most candidates earned partial credit. In general, candidates did well at identifying meaningful reference points for setting capital requirements, while they had difficulty expressing the reference points as a $TV@R$ measurement.
Part a Candidates were expected to know: • Pros and cons of default avoidance as a capital requirement • That this requirement focuses on events in the tails of the distribution where the ERM model is least reliable and most poorly understood • That this requirement does not recognize that significant partial losses of capital are important to shareholders which would require the threshold capital level to be well above the relatively remote default avoidance level. The majority of candidates received partial credit or full credit. Common mistakes were: • Indicating that default avoidance produces an excessive capital threshold leaning towards overcapitalization • Identifying other meaningful reference points as drawbacks • Repeated answers between parts a & b
Part b The candidate was expected to know any two of several other reference points. A brief description of any two of many other reference points received full credit. Generally candidates did well on this part and frequently received full credit. A common error was to provide capital requirement metrics such as VAR, $TV@R$, $XTV@R$, or EPD as reference points.
Part c Candidates were expected to know how to express one of the capital requirements as a $TV@R$ measurement. They were expected to select a maximum capital loss tolerance and express this as

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a 1-in-y-years event, and identify the relationship of the TV@Rxx as a 1-in-y-year event, and thus express the minimum capital requirement as a multiple of TV@Rxx, or a capital requirement plus TV@Rxx as a buffer.

Common errors included:

- Defining TV@R instead of providing the requested response
- Not using TV@R in the response
- Indicating TV@R as the entire capital requirement instead of a buffer